

Augmented perspective covariance, Gibbs gap, and information boundary

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claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-092 closed the normalized Cartan estimate only for the regular mutually orthogonal strict-past no-revisit one-shot class. It left two proposed successors: a complete signed rational/linear near estimate, denoted H_N , and a progressive overlap/revisit assembly, denoted H_A . This note tests the strongest remaining shortcuts before another long packet expansion.

Four exact conclusions result.

First, every term in the R-092 augmented one-reveal perspective expression has a shorter unconditional normal form. For a symmetric reveal it becomes the covariance between the revealed coefficient and the centered quadratic increment. This identifies the sign mechanism without discarding the perspective square or counting the terminal energy twice.

Second, the complete four-row production coefficient admits a bounded smooth same-root Gaussian shift for which this augmented density is strictly negative for every derivative payment. The example is production-specific, but it is not a torus control and is not a paid H_N counterexample. A separate exact cutoff-two theorem proves that the full 9/20 Cameron–Martin payment dominates every amplitude scaling of every shell-two control. Thus the local sign example diagnoses covariance, while its attempted paid scaling is closed negatively as a route to falsification.

Third, conditioning on the same-root coefficient is not free. The missing entropy is the mutual information between the Gaussian reveal and the coefficient. It is $\log N$ for N equiprobable coefficient bins and is infinite for the smooth non-atomic deterministic production reveal used below. Hence the coefficient-conditioned R-092 positivity criterion cannot be inserted into the unconditional A9 entropy argument without paying a new information term.

Fourth, triangular entropy disintegration has an exact Gibbs-gap form. At $q = 10/9$, both nonnegative gaps carry coefficient 9/10, while the source energy carries 9/20. After taking the directed union of all temporally faithful finite source refinements, R-081 temporal factorisation and R-087 CORE show that the infimum is exactly the Nelson free energy. This statement is false for a fixed source chart. Moreover near minimisers make both the Gibbs relative entropy and the fibre entropy tend to zero. Fibre entropy is therefore exact bookkeeping, not a uniformly positive reserve with which to pay H_N .

These conclusions materially shorten the roadmap. The source-coordinate inequality called (11.8) in R-092 is full source OVERLAP and is already equivalent to Nelson over the refinement union. If the name H_A is retained, it must mean only the assembly defect required to derive that inequality from the exact R-079 packet. The surviving analytic choices are a coefficient- unconditioned root-local paid production estimate or a direct static $q = 10/9$ Nelson estimate. Neither is proved here. Complete uniform H_N , full OVERLAP, Nelson, floor removal, an interacting measure, and Sector A remain open. Tier stays T4.

The load-bearing authorities are R-079 for the nonduplicating endpoint, R-081 for strict-past temporal factorisation, R-087 for fixed-cutoff variational CORE, R-091 for the complete four-row scalar production coefficient, and R-092 for the augmented perspective and triangular entropy identities. The cutoff-two

theorem additionally uses the A7 exact renormalised counterterm and the A12 Pauli–Fierz operator bound $0 \preceq B(z) \preceq \beta_{\text{op}}|z|^2 I_6$.

2 The unconditional augmented normal form

Let $\mathcal{F}$ be the past. Write $z = G_0 + g$, where G_0 is $\mathcal{F}$ -measurable,

$$\mathbb{E}_{\mathcal{F}}g = 0, \quad \Gamma = \mathbb{E}_{\mathcal{F}}(g \otimes g). \quad (2.1)$$

Let $B = B^T \succeq 0$ be revealed at the new level and let $R = R^T \succ 0$ be $\mathcal{F}$ -measurable. Define

$$A = B + 2R, \quad \bar{B} = \mathbb{E}_{\mathcal{F}}B, \quad \bar{A} = \bar{B} + 2R, \quad (2.2)$$

$$q_B = \mathbb{E}_{\mathcal{F}}(Bz), \quad m = A^{-1}Bz, \quad m_0 = \bar{A}^{-1}q_B, \quad (2.3)$$

and

$$\Theta_R(B) = B - B(B + 2R)^{-1}B, \quad Y = \Theta_R(B)^{1/2}z, \quad r = \mathbb{E}_{\mathcal{F}}Y. \quad (2.4)$$

The principal square root is well-defined even when B and R do not commute. Indeed,

$$\begin{pmatrix} B & B \\ B & B + 2R \end{pmatrix} \succeq 0, \quad (x, y) \mapsto (x + y)^T B(x + y) + 2y^T R y,$$

so the Schur complement of $B + 2R \succ 0$ gives $\Theta_R(B) \succeq 0$; also $\Theta_R(B) \preceq B$ because $B(B + 2R)^{-1}B \succeq 0$. All spaces in this theorem are finite-dimensional. Assume the displayed conditional moments are finite, in particular $\mathbb{E}_{\mathcal{F}}\|B\|$, $\mathbb{E}_{\mathcal{F}}z^T B z$, $\mathbb{E}_{\mathcal{F}}\|Bz\|$, and $\mathbb{E}_{\mathcal{F}}\|Y\|^2$; hence $\bar{B}$, q_B , r and every term below are well-defined. The production fixtures satisfy these hypotheses.

Theorem 2.1 (complete augmented one-reveal normal form)

The complete R-092 augmented density equals

$$\begin{aligned} \mathfrak{D}_R &:= \mathbb{E}_{\mathcal{F}} \left[(m - m_0)^T A(m - m_0) + |Y - r|^2 \right] - \bar{B} : \Gamma \\ &= \mathbb{E}_{\mathcal{F}}(z^T B z) - \bar{B} : \Gamma - q_B^T \bar{A}^{-1} q_B - |r|^2. \end{aligned} \quad (2.5)$$

No endpoint square is duplicated in this identity.

Proof. Since $Am = Bz$ and $\mathbb{E}_{\mathcal{F}}(Am) = q_B = \bar{A}m_0$,

$$\mathbb{E}_{\mathcal{F}}(m - m_0)^T A(m - m_0) = \mathbb{E}_{\mathcal{F}}z^T B A^{-1} B z - q_B^T \bar{A}^{-1} q_B. \quad (2.6)$$

Also

$$\mathbb{E}_{\mathcal{F}}|Y - r|^2 = \mathbb{E}_{\mathcal{F}}z^T \Theta_R(B) z - |r|^2. \quad (2.7)$$

The pointwise matrix identity

$$B(B + 2R)^{-1}B + \Theta_R(B) = B \quad (2.8)$$

then gives (2.5). $\square$

The theorem is the unconditional compression of R-092 (9.11). It keeps the Schur square, the terminal Θ_R martingale energy, and the covariance subtraction together. In particular no conclusion can be drawn from the sign of any one of them separately.

Corollary 2.2 (even-reveal covariance classifier)

Assume $G_0 = 0$, the conditional law of g is centrally symmetric, and $B(-g) = B(g)$. Then $q_B = r = 0$, and

$$\mathfrak{D}_R = \mathbb{E}_{\mathcal{F}}[B(g) : (g \otimes g - \Gamma)]. \quad (2.9)$$

For an independent copy g' this can also be written

$$\mathfrak{D}_R = \frac{1}{2} \mathbb{E}_{\mathcal{F}}[(B(g) - B(g')) : (g \otimes g - g' \otimes g')]. \quad (2.10)$$

In one dimension, if $B = \beta(g^2)$, then $\mathfrak{D}_R$ is nonnegative when β is increasing and nonpositive when β is decreasing. The sign is strict when g^2 is nondegenerate and β is strictly monotone on its essential range; equivalently, for an independent g' , the product $(\beta(g^2) - \beta(g'^2))(g^2 - g'^2)$ is nonzero with positive probability. The classification is independent of R , but only for the *sum* in (2.5), not term by term.

3 The exact production-coefficient sign fixture

On the pure-doublet scalar ray the R-091 complete four-row coefficient is

$$b(u) = 4c_0u^2 + 4c_1 \left(u - \frac{5}{9} \frac{u^3}{u^2 + e} \right)^2, \quad c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}, \quad (3.1)$$

where $e > 0$ is the density floor and $P = 4 + 10^{-12}$. Put $u = \sqrt{e}t$ and

$$\varrho(t) = t - \frac{5}{9} \frac{t^3}{1 + t^2} = \frac{t(4t^2 + 9)}{9(1 + t^2)}. \quad (3.2)$$

Then

$$\varrho'(t) = \frac{4t^4 + 3t^2 + 9}{9(1 + t^2)^2} > 0, \quad (3.3)$$

and

$$\frac{d}{dt}b(\sqrt{e}t) = 8e\{c_0t + c_1\varrho(t)\varrho'(t)\} > 0 \quad (t > 0). \quad (3.4)$$

Thus the restriction of b to the positive interval $[\sqrt{e}/2, 3\sqrt{e}/2]$ has a smooth inverse. Direct evaluation gives

$$b_- := b(\sqrt{e}/2) = \frac{9e}{250P}, \quad b_+ := b(3\sqrt{e}/2) = \frac{8937e}{42250P}, \quad (3.5)$$

$$\Delta := b_+ - b_- = \frac{3708e}{21125P} > 0. \quad (3.6)$$

Define the decreasing smooth function

$$\beta(s) = b_- + \frac{\Delta}{1 + s}, \quad s \geq 0, \quad (3.7)$$

and, crucially, use the positive-branch inverse

$$u(s) = \left(b|_{[\sqrt{e}/2, 3\sqrt{e}/2]} \right)^{-1}(\beta(s)), \quad a(g) = u(g^2) - \sqrt{e}. \quad (3.8)$$

The global even function b has no global inverse; (3.8) is the correct definition. The shift a is bounded, smooth, and even, and its exact production coefficient is $B(g) = \beta(g^2)$.

For a standard Gaussian G , Corollary 2.2 gives, for every $R > 0$,

$$\begin{aligned}\mathfrak{D}_R &= \Delta \mathbb{E} \frac{G^2 - 1}{1 + G^2} \\ &= \Delta \left\{ 1 - 2\mathbb{E} \frac{1}{1 + G^2} \right\} < 0.\end{aligned}\tag{3.9}$$

The strict inequality follows from strict Jensen because $s \mapsto (1 + s)^{-1}$ is strictly convex and $\mathbb{E}G^2 = 1$. The exact integral is

$$\mathbb{E} \frac{1}{1 + G^2} = \sqrt{\frac{\pi}{2}} e^{1/2} \operatorname{erfc}(1/\sqrt{2}) = 0.655679542418798 \dots\tag{3.10}$$

An exact rational non-Gaussian audit makes the cancellation transparent. Let $g = 0$ with probability $1/2$ and $g = \pm\sqrt{2}$ with probability $1/4$ each. Then $\Gamma = 1$,

$$B(0) = b_+, \quad B(\pm\sqrt{2}) = b_- + \frac{\Delta}{3} = \frac{3993e}{42250P},\tag{3.11}$$

$$\mathbb{E}B = \frac{1293e}{8450P},\tag{3.12}$$

and the complete augmented density is

$$\boxed{\mathfrak{D}_R = -\frac{1236e}{21125P} < 0}, \quad \frac{1}{2}\mathfrak{D}_R = -\frac{618e}{21125P}.\tag{3.13}$$

For every $\eta > 0$, the perspective square and terminal contribution obey pointwise

$$(B + 2\eta) \left(\frac{Bg}{B + 2\eta} \right)^2 + \left(B - \frac{B^2}{B + 2\eta} \right) g^2 = Bg^2,\tag{3.14}$$

which proves the η -independence without deleting either term.

Equations (3.9)–(3.14) sharpen the mechanism under the existing R-092 perspective-positivity boundary. They are a same-root scalar-coordinate fixture. There is no torus realization $a = T_J v$, no spatial shell, no $c = Da$, and no low, paid, heat, or forest accounting. They do not refute a complete paid H_N estimate.

4 The smallest genuine paid lift is coercive

The failed attempt to amplify the preceding local sign can be closed exactly at the first genuine two-shell cutoff. Let $L = 16$, $\omega = 2\pi/L$, and

$$A(n) = \{r + Z\omega^2|n|^2 + Y\omega^4|n|^4\}I_3 + \operatorname{diag}(m_1, m_2, m_3) + k_{\text{lock}}(I - P_0)\tag{4.1}$$

be the hash-pinned A1 precision symbol. Let X_2 be the canonical six-real Gaussian with $\|n\|_\infty \leq 2$. For every bounded smooth $\mathcal{F}_1$ -measurable source v supported in shell $\|n\|_\infty = 2$, put $a = T_2 v$. The full endpoint action is

$$I_2(v) = \mathbb{E} \left[V_2^{\text{ren}}(X_2 + a) + \frac{3}{20} \|X_2 + a\|_6^6 + \frac{9}{20} \|v\|^2 \right].\tag{4.2}$$

Because (4.2) uses the unexpanded terminal functional, the complete six-real coefficient, current square, covariance trace, low endpoint, paid subtraction, heat transport, and R-063 forest occur exactly once.

The A1 Pauli–Fierz bound is

$$0 \preceq B(z) \preceq \beta_{\text{op}} |z|^2 I_6, \quad \beta_{\text{op}} = 0.0423749999999894 \dots\tag{4.3}$$

Dropping the nonnegative current square gives

$$V_2^{\text{ren}}(z) \geq -c_2 \|z\|_2^2, \quad c_2 = \frac{\beta_{\text{op}}}{2} \sum_{i=1}^3 \text{tr } \Gamma_{2,i}. \quad (4.4)$$

Direct finite symbol summation, with the canonical complex-to-six-real factor two, gives

$$\sum_i \text{tr } \Gamma_{2,i} = \frac{2}{L^3} \sum_{\|n\|_\infty \leq 2} \omega^2 |n|^2 \text{tr } A(n)^{-1} = 0.226418249643189 \dots, \quad (4.5)$$

$$c_2 = 0.00479723666431388 \dots \quad (4.6)$$

Exhaustive shell enumeration gives

$$\lambda_{2,\min} := \min_{\|n\|_\infty=2} \lambda_{\min} A(n) = 0.311897688710642 \dots, \quad (4.7)$$

and hence the conservative real-coordinate bound

$$\mathbb{E} \|T_2 v\|_2^2 \leq \Lambda_2 \mathbb{E} \|v\|^2, \quad \Lambda_2 = \frac{2}{\lambda_{2,\min}} = 6.41235915619583 \dots \quad (4.8)$$

The same symbol sum gives

$$\mathbb{E} \|X_2\|_2^2 = 2 \sum_{\|n\|_\infty \leq 2} \text{tr } A(n)^{-1} = 1236.11554774495 \dots \quad (4.9)$$

Using $\|X_2 + a\|_2^2 \leq 2\|X_2\|_2^2 + 2\|a\|_2^2$ and discarding the nonnegative sextic yields the finite-cutoff theorem

$$\boxed{I_2(v) \geq 0.388476791102297 \dots \mathbb{E} \|v\|^2 - 11.859877653941 \dots} \quad (4.10)$$

In particular $I_2(\lambda v) \rightarrow +\infty$ as $|\lambda| \rightarrow \infty$ for every nonzero v . No same-root selector or amplitude scaling of the local fixture gives an unbounded negative paid two-shell packet. This fires NG-2026-07-27-A13-LOCAL-PERSPECTIVE-PAID-SCALING.

The theorem is finite-cutoff only. Its crude c_J uses the total covariance trace and grows with J . It does not prove a cutoff-uniform H_N estimate.

5 Coefficient reveal has an information price

Let G have law γ and let $B = f(G)$ be a same-root coefficient reveal. Disintegration gives

$$\mathbb{E}_B H(\mathcal{L}(G \mid B) \mid \gamma) = I(G; B). \quad (5.1)$$

Thus a proof that first conditions on B , applies the R-092 coefficient-conditioned moment-matched criterion, and then reuses the unconditional Gaussian entropy must pay $I(G; B)$.

For N equiprobable bins of B ,

$$I(G; B_N) = H(B_N) = \log N, \quad (5.2)$$

even when the Cameron–Martin control is zero. For the production reveal $B = \beta(G^2)$ in (3.7), β is strictly monotone and B determines G^2 . Its law is non-atomic. Quantising into N equiprobable bins and using data processing gives

$$I(G; B) \geq I(G; B_N) = \log N \quad \text{for every } N, \quad (5.3)$$

so

$$\boxed{I(G; B) = +\infty.} \quad (5.4)$$

Equivalently, the joint law of $(G, f(G))$ lies on a graph and is singular with respect to the product of its marginals when the reveal is non-atomic.

This is a method no-go, not an H_N counterexample. A coefficient-unconditioned martingale proof may still control the covariance in (2.9). It fires NG-2026-07-27-A13-COEFFICIENT-REVEAL-FREE-CONDITIONING.

6 Exact Gibbs gap in source coordinates

Let $\gamma = \bigotimes_k \gamma_k$ be a finite standard Gaussian source, $S_\pi(\xi) = \sum_k S_k \xi_k$, and $\mu_C = (S_\pi)_\# \gamma$. Let

$$T_h(\xi) = \xi + h(\xi), \quad h_k = h_k(\xi_1, \dots, \xi_{k-1}) \quad (6.1)$$

be bounded, smooth, and strictly triangular. Put $\rho_h = (T_h)_\# \gamma$, $\nu_h = (S_\pi)_\# \rho_h$, and let Φ_h be the R-092 fibre relative entropy. Strict triangularity makes T_h recursively invertible with determinant one, so

$$\frac{1}{2} \mathbb{E} \sum_k \|h_k\|^2 = H(\rho_h \mid \gamma) = H(\nu_h \mid \mu_C) + \Phi_h. \quad (6.2)$$

Fix a cutoff J . Throughout this section assume the R-087 CORE payoff hypotheses: G_J is real-valued, continuous, bounded below, and satisfies the local polynomial envelope R-087 (8.2). Thus, for $q > 0$, $0 < Z_J < \infty$, where

$$Z_J = \int e^{-qG_J} d\mu_C, \quad \nu_J^* = Z_J^{-1} e^{-qG_J} \mu_C. \quad (6.3)$$

All charts below are bounded smooth cylindrical-simple triangular charts, so their source energy, payoff expectation, and displayed relative entropies are finite. The identities extend only when both sides are well-defined; no undefined $\infty - \infty$ subtraction is intended.

Theorem 6.1 (Gibbs-gap identity)

For every source chart and every strictly triangular h ,

$$\boxed{\begin{aligned} \mathcal{A}_J(h) &:= \mathbb{E} G_J(S_\pi(\xi + h)) + \frac{1}{2q} \mathbb{E} \sum_k \|h_k\|^2 \\ &= -\frac{1}{q} \log Z_J + \frac{1}{q} H(\nu_h \mid \nu_J^*) + \frac{1}{q} \Phi_h. \end{aligned}} \quad (6.4)$$

Proof. The definition of ν_J^* gives

$$H(\nu_h \mid \nu_J^*) = H(\nu_h \mid \mu_C) + q \mathbb{E}_{\nu_h} G_J + \log Z_J. \quad (6.5)$$

Substitute (6.2) into the left side and rearrange. □

At the A13 value $q = 10/9$,

$$\frac{1}{2q} = \frac{9}{20}, \quad \frac{1}{q} = \frac{9}{10}. \quad (6.6)$$

The two gap terms therefore carry 9/10, not 9/20.

A fixed chart does not attain the Gibbs infimum

Take one standard source block, $S = 1$, and $G(x) = x^2/2$. Strict triangularity forces h to be a constant translation. Hence

$$\inf_h \mathcal{A}(h) = \frac{1}{2}, \quad (6.7)$$

whereas

$$-\frac{1}{q} \log \mathbb{E} e^{-qG(G_0)} = \frac{1}{2q} \log(1+q) = \frac{9}{20} \log \frac{19}{9} < \frac{1}{2}. \quad (6.8)$$

Thus no equality may be claimed for each fixed decomposition. This fires
NG-2026-07-27-A13-FIXED-SOURCE-CHART-GIBBS-ATTAINMENT.

The temporally faithful refinement union is exact

Let $\mathfrak{T}_J$ be the directed union over all finite temporal Gaussian source refinements which retain every past Brownian evaluation used by a bounded smooth cylindrical-simple control, including kernel or auxiliary coordinates when the terminal physical map forgets them. The physical terminal map and source cost use the canonical covariance normalisation.

Theorem 6.2 (source-union/CORE equality)

At each fixed cutoff,

$$\inf_{(S_\pi, h) \in \mathfrak{T}_J} \mathcal{A}_J(h) = -\frac{1}{q} \log \mathbb{E}_{\mu_C} e^{-qG_J}. \quad (6.9)$$

Proof. The lower bound follows from (6.4). For the reverse bound, choose an ε -minimising bounded smooth cylindrical-simple progressive drift from R-087 CORE. Its finitely many past Brownian evaluations and terminal Gaussian image define a chart in $\mathfrak{T}_J$. On every strict-past time interval, R-081 Douglas factorisation gives the same terminal displacement and a source shift whose squared norm is no larger than the original interval control cost. The auxiliary coordinates retain the information required to make every shift a function of earlier source blocks. Therefore the source action is no larger than the chosen Boué–Dupuis action. Letting $\varepsilon \downarrow 0$ proves the reverse inequality. $\square$

This theorem is a coordinate equivalence, not a lower bound uniform in J . It corrects the fixed-chart ambiguity in R-092 (11.8).

Corollary 6.3 (near-minimiser rigidity)

If $\mathcal{A}_J(h_n)$ tends to the value in (6.9), then

$$H(\nu_{h_n} \mid \nu_J^*) + \Phi_{h_n} \longrightarrow 0. \quad (6.10)$$

Both nonnegative terms tend to zero separately. Quantitatively, if the action gap is δ_n , then

$$H(\nu_{h_n} \mid \nu_J^*) \leq q\delta_n, \quad \Phi_{h_n} \leq q\delta_n, \quad (6.11)$$

and Pinsker gives

$$\|\nu_{h_n} - \nu_J^*\|_{\text{TV}} \leq \sqrt{q\delta_n/2} = \sqrt{5\delta_n/9}. \quad (6.12)$$

Consequently no proof can assume a uniform positive fibre-entropy lower bound on every dangerous near-minimising branch. This does not preclude using the actual, possibly vanishing, Φ_h inside a coupled estimate. This fires NG-2026-07-27-A13-FIBRE-ENTROPY-UNIFORM-RESERVE.

7 Causal geometry does not supply the missing reserve

There is no filtration-preserving orthogonal QR compression of revisited physical ranges. Indeed, a finite block-lower-triangular orthogonal operator Q has a lower-triangular inverse, while $Q^{-1} = Q^T$ is block upper triangular. Hence Q^{-1} and Q are block diagonal. Induction over the diagonal blocks proves

$$\boxed{Q \text{ block lower triangular and orthogonal}} \implies \boxed{Q \text{ block diagonal}}. \quad (7.1)$$

Thus a causal orthogonal change of coordinates cannot coalesce repeated ranges across time. This fires NG-2026-07-27-A13-CAUSAL-ORTHOGONAL-QR.

Two exact entropy allocations show why levelwise physical accounting is not stable. For $S_1 = S_2 = 1/\sqrt{2}$ and $h = (t, 0)$, the source relative entropy is $t^2/2$, the final physical entropy is $t^2/4$, and the fibre

entropy is $t^2/4$. The physical entropy of the first prefix alone is $t^2/2$ and then drops after the second independent range is added.

For the strictly triangular feedback $h = (0, 2\xi_1)$, the source entropy is 2. The terminal physical Gaussian has variance 5, so

$$H(\nu_h \mid \mu_C) = 2 - \frac{1}{2} \log 5, \quad \Phi_h = \frac{1}{2} \log 5. \quad (7.2)$$

The split is exact but does not equal a fixed multiple of a polar-kernel norm. At the other extreme, a minimal deterministic source shift can have $\Phi_h = 0$. These examples are consistent with (6.10): fibre entropy stores some revisit loops but is not coercive on the full variational class.

The semantic consequence is important. The R-092 source action

$$\mathbb{E} G_J(S_\pi(\xi + h)) + \frac{9}{20} \mathbb{E} \sum_k \|h_k\|^2 \quad (7.3)$$

over $\mathfrak{T}_J$ is full source OVERLAP, denoted $\text{OVERLAP}_{\text{src}}$ here. By Theorem 6.2 its uniform lower bound is the $q = 10/9$ Nelson bound itself. It is not a weaker independent H_A module. The label H_A may still be used for the packet-assembly theorem which derives (7.3) from R-079 without loss or duplication.

8 Surviving route and root-local scaling

The finite-cutoff estimate (4.10) shows that payment strength is not the local obstruction. Its failure to be uniform comes from bounding the entire base covariance trace before the exact heat/low/forest cancellation. The next production-specific target must keep that cancellation root-local.

At high root j , the order-four A1 precision gives

$$\text{tr } \Delta \Gamma_j \asymp 2^j. \quad (8.1)$$

For a future control shell $k > j$, the terminal covariance map has

$$\|C_k\|_{\text{op}} \asymp 2^{-4k}. \quad (8.2)$$

Their control-dependent product therefore has the favourable prototype

$$2^{j-4k}, \quad k > j, \quad (8.3)$$

whose triangular double sum is strongly convergent. This is a scaling diagnostic, not yet a theorem: the coefficient secant, low object, backward heat, paid subtraction, and R-063 forest must be placed into the same root-local identity before (8.3) may be used.

The exact next alternatives are therefore:

1. prove a coefficient-unconditioned root-local paid production estimate for the full augmented H_N packet, cancelling the cutoff-growing base term before applying (8.3); or
2. prove the static uniform free-energy bound

$$-\frac{9}{10} \log \mathbb{E} e^{-(10/9)G_J} \geq -C \quad (8.4)$$

directly, using an enhanced-model or integrated A9 smart-path estimate.

By Theorem 6.2 the second alternative is not a shortcut around OVERLAP; it is exactly Nelson. The first alternative remains a sufficient internal route to it. Conditioning on B , fixed-chart optimisation, an independent uniform fibre reserve, causal orthogonal QR, and amplification of the local perspective sign are now retired. Coupled use of the actual fibre term is not excluded.

9 Enhanced-model polynomial BG criterion

There is one exact static sufficient criterion worth retaining. Write

$$G_J(z) = V_J^{\text{ren}}(z) + \varepsilon_* \|z\|_6^6, \quad \varepsilon_* = \frac{3}{20}, \quad (9.1)$$

and, for $h = T_J v$,

$$E_v = \int |v_t|^2 dt, \quad Y_v = \|X_J + h\|_6^6. \quad (9.2)$$

Assume that an explicit finite or countable A7 model family $Z_{\tau,J} \geq 0$ and exponents $a_\tau, b_\tau \geq 0$ satisfy, for every finite-cutoff shift,

$$-\{V_J^{\text{ren}}(X_J + h) - V_J^{\text{ren}}(X_J)\} \leq C_0 + \sum_{\tau} Z_{\tau,J} E_v^{a_\tau} Y_v^{b_\tau}, \quad (9.3)$$

where C_0 is independent of J and h , and assume

$$s_\tau := 1 - a_\tau - b_\tau > 0, \quad \inf_J \mathbb{E} V_J^{\text{ren}}(X_J) > -\infty. \quad (9.4)$$

Multifactor Young gives, for arbitrary positive allocations,

$$Z E^a Y^b \leq \eta E + \zeta Y + C_{\eta,\zeta} Z^{1/(1-a-b)}. \quad (9.5)$$

If the allocations also obey

$$\sum_{\tau} \eta_{\tau} < \frac{9}{20}, \quad \sum_{\tau} \zeta_{\tau} < \frac{3}{20}, \quad \sum_{\tau} C_{\eta_{\tau}, \zeta_{\tau}} \sup_J \mathbb{E} Z_{\tau,J}^{1/s_{\tau}} < \infty, \quad (9.6)$$

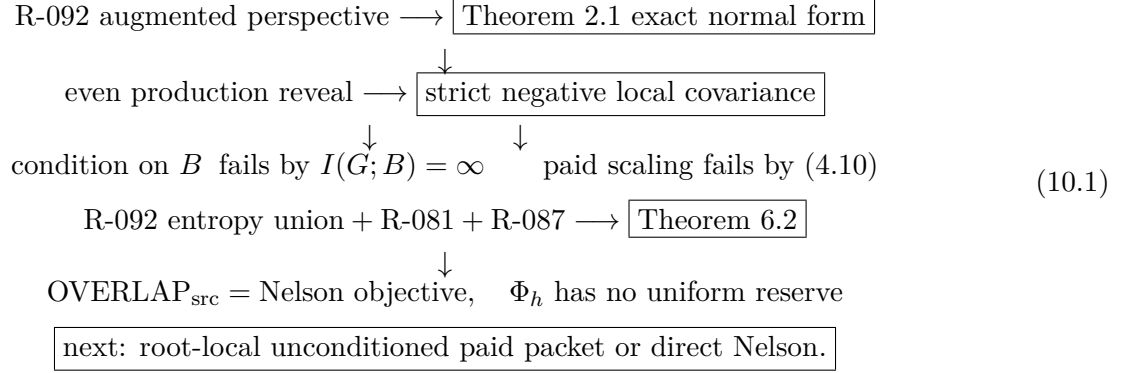
then (9.3)–(9.6), followed by R-087 CORE, prove the uniform $q = 10/9$ Nelson estimate. For a finite family the last condition reduces to the corresponding finitely many uniform moments. For an infinite family, individual moment bounds without the displayed weighted summability are not enough.

For reference, the following is only the arithmetic ledger obtained by substituting the displayed candidate exponent pairs from the cited predecessor notes into $s = 1 - a - b$. It is not a new transfer theorem or an authority for (9.3): every row must still be reconstructed inside the exact signed shifted packet before the criterion applies.

Family	a	b	$s = 1 - a - b$	moment $1/s$
Q against h^2 , $\kappa = 1/10$	21/40	19/120	19/60	60/19
R-076 sharp base-frozen cubic	2/5	8/15	1/15	15
comparable rational cubic	11/20	19/60	2/15	15/2
rational Q branch	11/20	3/20	3/10	10/3
G /Hessian branch	2/5	11/30	7/30	30/7
high- U branch	11/40	49/120	19/60	60/19
R-092 worst regular control	4/5	1/6	1/30	30
R-092 regular root	3/5	1/3	1/15	15
coarse critical current	3/4	1/4	0	not covered
coarse graph term	1/2	1/2	0	not covered

The largest finite required moment in this finite inherited ledger is 30, already available for the unshifted A7 forests. The missing input is the complete shifted signed reconstruction (9.3), including a J - and h -uniform C_0 and, for any infinite packet family, the summed remainder condition in (9.6). A7 already supplies the exact base centering $\mathbb{E} V_J^{\text{ren}}(X_J) = 0$, hence the lower bound in (9.4), for the actual model. Extracting a translated model norm pathwise is forbidden by the R-087 rare-event example. Exponentiating that norm is worse: homogeneous third- and fourth-chaos tails do not provide the exponential moments of their fifteenth or thirtieth powers. Thus the expectation must remain inside the exact Cartan remainders. The criterion is noncircular, but no current authority proves its hypothesis.

10 Proof and failure evidence map



The route decisions are:

Route	Verdict	Reason / successor
Separate augmented positivity	failed	Exact production covariance is negative for every R ; keep the complete paid packet.
Amplify the local fixture	failed	The cutoff-two full action satisfies (4.10); local amplitude scaling is coercive.
Condition on B for free	failed	The missing mutual information is unbounded and is infinite for the smooth reveal.
Optimise one source chart	failed	The quadratic one-block gap (6.7)–(6.8) is strict.
Uniform fibre reserve	failed	Near minimisers force $\Phi_h \rightarrow 0$.
Causal orthogonal QR	failed	Lower-triangular orthogonal maps are block diagonal.
Temporal refinement union	exact	It reproduces CORE and hence the Nelson free energy, without proving a uniform bound.
Root-local paid production	open / favoured	Cancel the base heat/forest term first; then exploit the summable scale (8.3).

The new formal failure authorities are

- AUDIT-2026-07-27-A13-R092-AUGMENTED-PRODUCTION-COVARIANCE;
- NG-2026-07-27-A13-LOCAL-PERSPECTIVE-PAID-SCALING;
- NG-2026-07-27-A13-COEFFICIENT-REVEAL-FREE-CONDITIONING;
- NG-2026-07-27-A13-FIXED-SOURCE-CHART-GIBBS-ATTAINMENT;
- NG-2026-07-27-A13-FIBRE-ENTROPY-UNIFORM-RESERVE;
- NG-2026-07-27-A13-CAUSAL-ORTHOGONAL-QR.

11 Devil’s-advocate review

1. **Objection:** The negative density (3.9) is already a paid production counterexample. **UPHELD AGAINST THAT CLAIM.** It is only a scalar same-root coordinate fixture. The genuine cutoff-two paid lift instead satisfies the positive coercive margin (4.10).
2. **Objection:** The inverse in (3.8) is ill-defined because b is even. **VALID WITH REPAIR.** The note uses only the strictly increasing positive branch $b|_{[\sqrt{e}/2, 3\sqrt{e}/2]}$; no global inverse is asserted.

3. **Objection:** The perspective payment R could restore the sign. **DISMISSED FOR THE AUGMENTED LOCAL SUM.** Identity (3.14) proves exact R -independence after the terminal Θ_R energy is included once. Individual pieces remain R -dependent.
4. **Objection:** Mutual information merely reflects a poor discretisation. **DISMISSED.** Equiprobable quantisations give the lower bound $\log N$ for every N ; data processing then forces infinite information in the non-atomic deterministic reveal.
5. **Objection:** The source-infimum equality holds in the fixed R-092 chart. **DISMISSED.** The explicit quadratic one-block calculation gives a strict gap. Equality requires the directed union of temporally faithful refinements retaining the controller's past Gaussian information.
6. **Objection:** Fibre entropy has a uniform positive reserve which can independently pay the remaining signed near term. **UPHELD AS FALSE.** The exact Gibbs gap forces $\Phi_h \rightarrow 0$ on every variational near-minimising sequence. Coupled use of the actual vanishing fibre term is not excluded.
7. **Objection:** The finite-cutoff bound proves uniform H_N . **UPHELD AS FALSE.** Its crude constant contains the total Γ_J trace. Uniformity requires the root-local heat/low/forest cancellation before summation.
8. **Objection:** The scale 2^{j-4k} closes the next theorem. **UPHELD AS PREMATURE.** It is only the favourable order-counting of the covariance/control pair. The full coefficient secant and exact nonduplication ledger remain to be proved.
9. **Objection:** A six-real or source-normalisation factor is hidden in the cutoff-two constants. **DISMISSED WITH EXECUTABLE CROSS-CHECKS.** The gradient trace uses $2/L^3$, the integrated field trace uses the real-pairing factor 2, and the source map uses $\Lambda_2 = 2/\lambda_{2,\min}$. Primary physical-coordinate summation and independent spectral summation agree on all eight derived constants and on the four defining identities. All quantities are evaluated in the same finite-torus action convention; no unrecorded unit conversion is used.
10. **Objection:** Literal output constants could mask a wrong derivation. **VALID WITH REPAIR.** Every reported cutoff-two number and density is recomputed from the pinned A1 inputs. Exact rationals occur only as labelled test oracles; the JSON payload stores the computed expressions, and the integrated verifier checks their algebraic relations and an independently implemented spectrum.
11. **Objection:** The $J = 2$ enumeration supports a convergence or continuum claim. **UPHELD AS FALSE.** It is an exhaustive finite enumeration only. No cutoff extrapolation is made, and the total-field covariance in its lower constant is precisely why (4.10) is not uniform.
12. **Objection:** Individual model moments make the countable BG sum finite. **UPHELD AS FALSE.** Criterion (9.6) explicitly requires the weighted sum of Young remainders. A7 supplies exact base centering and the finite listed moments, not the missing shifted reconstruction, uniform C_0 , or infinite-family summability.

External review should concentrate on the source-union realisation in Theorem 6.2, the six-real normalisation in (4.5)–(4.9), and the root-local placement required before the scaling (8.3) is used.

12 Result footer

Result ID:	A13-CLASSII-AUGMENTED-PERSPECTIVE-GIBBS-GAP- INFORMATION-BOUNDARY
Ledger:	R-093.
Precise statement:	The complete augmented one-reveal density has the

exact unconditional normal form (2.5); a centered symmetric even reveal reduces it to the covariance (2.9). The exact production coefficient gives a bounded smooth negative local fixture, while the genuine J=2 paid action is coercive on bounded smooth predictable shell-two sources. Coefficient revelation costs mutual information. Under R-087 hypotheses, the temporally faithful directed source union equals CORE and has the exact Gibbs gap (6.4).

Scope: Finite-dimensional augmented identities; fixed-floor production sign diagnostic; canonical A1 cutoff J=2; bounded smooth predictable shell-two sources; and fixed-cutoff source refinements with the R-087 payoff and finite-entropy hypotheses. The BG statement is a conditional criterion requiring uniform shifted reconstruction and weighted remainder summability.

Dependencies: A1; A7; A12; R-079; R-081; R-087; R-091; R-092.

Evidence grade: ANALYTIC, EXACT, EXECUTED, BOUNDARY.

Reproduction command: python codes/foundations/a13_classii_augmented_perspective_gibbs_gap_information_boundary_verify.py

Expected output: Primary 71/71; independent 34/34; integrated 288/288; aggregate 393/393; exit zero.

Negative results: AUDIT-2026-07-27-A13-R092-AUGMENTED-PRODUCTION-COVARIANCE; NG-2026-07-27-A13-LOCAL-PERSPECTIVE-PAID-SCALING; NG-2026-07-27-A13-COEFFICIENT-REVEAL-FREE-CONDITIONING; NG-2026-07-27-A13-FIXED-SOURCE-CHART-GIBBS-ATTAINMENT; NG-2026-07-27-A13-FIBRE-ENTROPY-UNIFORM-RESERVE; NG-2026-07-27-A13-CAUSAL-ORTHOGONAL-QR.

Falsification gate: Failure of the Schur identity or positivity, normal form, covariance classifier, exact production values, any of the eight cutoff-two constants or four defining identities, information/Gibbs/source-union formulae, authority hashes, independent audit, PDF checks, exploration/registry links, counts, or scope firewall.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: No paid torus H_N counterexample, cutoff-uniform H_N, exact H_A assembly, OVERLAP_{src}, Nelson estimate, cutoff or floor removal, interacting measure, Sector-A closure, or T5--T7 claim is established.

Next required action: Prove the coefficient-unconditioned root-local H_N bound after heat/low/forest cancellation, then assemble complete packets losslessly into OVERLAP_{src}; or prove the equivalent static q=10/9 Nelson estimate directly.